

BCH 216 | COMPREHENSIVE TEST PREPARATION

Shuffled Practice Workbook · 123 Questions · 2025/2026 Session

Instructions: This workbook contains all test and practice questions shuffled together. Fill in all blanks with the most accurate answer from your lecture notes. The topic tag beneath each question indicates which lecture it is drawn from. The complete answer key is provided after the last question.

QUESTIONS

1. At lower temperature, melted strands of DNA as a result of denaturation are re-associated in a process called _____.

[Nucleic Acid Chemistry]

2. Heme structure includes a _____ ring bound as a tetradentate ligand to a central _____ atom.

[Heme Degradation]

3. An example of a biotransformation reaction harmful to the body is the conversion of _____ to vinyl chloride epoxide, which binds to DNA and RNA, leading to liver cancer.

[Xenobiotic Metabolism]

4. _____ hormones are water-soluble and bind to surface receptors on target cells.

[Biochemistry of Hormones]

5. The three main types of RNA involved in protein synthesis are _____, _____, and _____.

[Central Dogma]

6. The scientist who discovered that in DNA the ratio of purines to pyrimidines is always 1 is _____.

[Nucleic Acid Chemistry]

7. Bilirubin travels in the bloodstream as a complex with _____ because it is largely _____.

[Heme Degradation]

8. The chief Phase I oxidative reaction in drug metabolism is _____, catalysed by _____.

[Xenobiotic Metabolism]

9. Give four examples of Phase II conjugation reactions.

[Xenobiotic Metabolism]

10. Peptide hormones cannot penetrate the lipid bilayer, so they bind to _____ on the target cell membrane, triggering production of _____.

[Biochemistry of Hormones]

11. _____, the cousin of ATP, acts as a second messenger to many cells.

[Biochemistry of Hormones]

12. The _____ enzyme unwinds the DNA helix at the replication fork.

[DNA Replication]

13. The replication fork is produced by enzymes called _____ which break the hydrogen bonds holding the DNA helix together.

[DNA Replication]

14. The hormone-receptor complex for steroid hormones binds to a specific DNA regulatory sequence called the _____.

[Biochemistry of Hormones]

15. All _____ are mutagens but not all mutagens are _____.

[DNA Repair & Mutations]

16. Recombinant DNA technology is also commonly referred to as _____.

[Recombinant DNA Technology]

17. tRNA molecules function as _____ to bring amino acids to the _____ during protein synthesis.

[Central Dogma]

18. The short arm of a chromosome is labeled _____ and the long arm is labeled _____.

[Genetics & Heredity]

19. The primary purpose of the biotransformation process is to convert the _____, _____, and _____ compound to a more polar, water-soluble form.

[Xenobiotic Metabolism]

20. Acetaminophen toxicity is treated using _____.

[Xenobiotic Metabolism]

21. The information needed to direct the synthesis of protein is contained in mRNA in the form of a _____.

[Central Dogma]

22. The DNA strand that serves as the template for mRNA synthesis is called the _____ strand.

[Central Dogma]

23. The genetic code is said to be _____ because one codon codes for only one amino acid, yet one amino acid can be encoded by more than one codon.

[Central Dogma]

24. Hormones are produced by special cells known as endocrine glands; these glands secrete their hormones directly into the _____.

[Biochemistry of Hormones]

25. Xeroderma Pigmentosum follows an autosomal _____ pattern of inheritance.

[DNA Repair & Mutations]

26. Haploid cells contain _____ set(s) of chromosomes; in humans, haploid cells have _____ chromosomes.

[Genetics & Heredity]

27. In eukaryotes, DNA is typically broken up into a number of very long linear pieces called _____.

[Nucleic Acid Chemistry]

28. DNA is predominantly present in the _____ and small amounts are also present in the _____.

[Nucleic Acid Chemistry]

29. Accumulation of excess bilirubin in the blood causing yellowing of the skin and eyeballs is called _____.

[Heme Degradation]

30. RT-PCR uses _____ rather than DNA as the starting template.

[Recombinant DNA Technology]

31. DNA replication is described as semi-conservative because _____.

[DNA Replication]

32. Enzyme _____ is defined as increased synthesis or decreased degradation of enzymes due to the presence of an exogenous substance.

[Xenobiotic Metabolism]

33. 20 cycles of PCR provide approximately _____ times amplification of the original DNA.

[Recombinant DNA Technology]

34. Phospholipase C breaks down PIP₂ into two second messenger products: _____ and _____.

[Biochemistry of Hormones]

35. The compound that gives urine its yellow colour is _____ while _____ gives faeces its red-brown colour.

[Heme Degradation]

36. The enzyme that synthesizes DNA from an RNA template is _____ and it is an RNA-dependent _____.

[Recombinant DNA Technology]

37. The main limitations of using E. coli in recombinant DNA technology are _____.

[Recombinant DNA Technology]

38. Ferrous iron released during heme degradation is sequestered by the protein _____ to reduce its toxicity.

[Heme Degradation]

39. Mutations are due to a change in the _____ of DNA and may result from faulty _____ or _____.

[DNA Repair & Mutations]

40. Characteristics of hormones include _____, _____, and _____.

[Biochemistry of Hormones]

41. An agent that increases the rate of mutation can also increase the probability of _____; thus all carcinogens are _____.

[DNA Repair & Mutations]

42. Cytochrome P450 enzymes absorb light at a wavelength of _____ nm when exposed to _____.

[Xenobiotic Metabolism]

43. Restriction enzymes are named after the _____ of the organism they are isolated from.

[Recombinant DNA Technology]

44. A gene library consisting only of expressed genes is called a _____ library.

[Recombinant DNA Technology]

45. During transcription, Thymine (T) in DNA is replaced by _____ in the mRNA transcript.

[Central Dogma]

46. Activation of G-protein by a hormone leads to dissociation of the _____ subunit, which then activates adenylate cyclase to generate _____.

[Biochemistry of Hormones]

47. The _____ enzyme adds nucleotides and performs proofreading and error correction.

[DNA Replication]

48. The synthesis of DNA molecules from an RNA template is done by _____.

[Central Dogma]

49. Nirenberg was awarded the Nobel Prize in _____ for deciphering the _____.

[Central Dogma]

50. The specific location in the genome where DNA replication begins is called the _____.

[DNA Replication]

51. The cycles in PCR are generally repeated by an automated machine called _____.

[Recombinant DNA Technology]

52. The principal organ of drug metabolism is the _____; other sites include _____ and _____.

[Xenobiotic Metabolism]

53. An individual with two identical alleles at a locus is _____; one with two different alleles is _____.

[Genetics & Heredity]

54. The term xenobiotic refers to chemical substances that are _____ to animal life.

[Xenobiotic Metabolism]

55. Cytosine + Deoxyribose + 2 phosphate groups = _____

[Nucleic Acid Chemistry]

56. The three major steps of the Central Dogma of Molecular Biology are _____, _____, and _____.

[Central Dogma]

57. Heme is a physiological substrate for _____.

[Heme Degradation]

58. Based on chemical structure, hormones can be classified into _____, _____, and _____.

[Biochemistry of Hormones]

59. Guanine + Ribose = _____; Adenine + Deoxyribose = _____.

[Nucleic Acid Chemistry]

60. Restriction enzymes belong to a larger class of enzymes called _____.

[Recombinant DNA Technology]

61. The specific location of a gene on a chromosome is called its _____.

[Genetics & Heredity]

62. DNA molecules which carry foreign DNA fragments to be cloned are known as _____.

[Recombinant DNA Technology]

63. Nucleotides are made up of _____, _____ and _____.

[Nucleic Acid Chemistry]

64. RNA molecules that act as enzymes are known as _____.

[Nucleic Acid Chemistry]

65. The gap or nick in DNA strands are sealed by _____.

[DNA Replication]

66. Why is bacteriophage preferred over plasmids when working with genomes of human cells?

[Recombinant DNA Technology]

67. The normal mechanism for repairing UV-induced DNA damage that is defective in XP is _____.

[DNA Repair & Mutations]

68. The universal currency of energy, ATP, is a derivative of _____.

[Nucleic Acid Chemistry]

69. A typical example of a Phase III reaction is further conjugation with _____.

[Xenobiotic Metabolism]

70. In DNA, Adenine pairs with Thymine by _____ hydrogen bonds, while Cytosine pairs with Guanine by _____ hydrogen bonds.

[Nucleic Acid Chemistry]

71. Retroviruses use _____ as their genetic material and synthesize DNA from it by a process known as _____.

[Central Dogma]

72. The genetic message coded by mRNA is translated by _____ into a protein.

[Central Dogma]

73. Uracil + Ribose = _____

[Nucleic Acid Chemistry]

74. A genetic disorder characterised by decreased ability to repair DNA damage caused by ultraviolet light is _____.

[DNA Repair & Mutations]

75. Plasmids, cosmids and bacteriophages are examples of _____ or _____.

[Recombinant DNA Technology]

76. The method of gene transfer based on the principle of high voltage electric pulses on plasma membranes is _____.

[Recombinant DNA Technology]

77. The pentose sugar found in RNA is _____ while the sugar found in DNA is _____.

[Nucleic Acid Chemistry]

78. The enzyme used to join two DNA fragments cut by the same restriction enzyme is _____.

[Recombinant DNA Technology]

79. The process in which many orally administered drugs are extensively metabolized in the liver before reaching systemic circulation is called the _____.

[Xenobiotic Metabolism]

80. The physical expression of an organism's genetic makeup is its _____, while the actual genetic constitution is its _____.

[Genetics & Heredity]

81. The presence of _____ in hemoglobin is responsible for the characteristic red pigmentation of blood.

[Heme Degradation]

82. Phase I reactions introduce functional groups such as _____, _____, or _____ into the parent compound.

[Xenobiotic Metabolism]

83. Cyclic AMP (cAMP) is formed from ATP by the action of the enzyme _____.

[Nucleic Acid Chemistry]

84. The enzyme that degrades heme is a _____ kDa heat shock protein containing _____ amino acid residues.

[Heme Degradation]

85. In PCR, new DNA strands are synthesized by _____.

[Recombinant DNA Technology]

86. The co-enzyme for all P450 enzymes is _____ and NOT _____.

[Xenobiotic Metabolism]

87. Prokaryotic DNA is not enclosed in a membrane; it is located in a specialized region called the _____.

[Nucleic Acid Chemistry]

88. Nucleic acids are polymers of _____ linked by _____, hence referred to as _____.

[Nucleic Acid Chemistry]

89. The universal red cell donor blood type is _____ while the universal red cell recipient blood type is _____.

[Genetics & Heredity]

90. The endogenous reactant for glucuronidation is _____ and the transferase is located in the _____.

[Xenobiotic Metabolism]

91. The process through which the genetic message in DNA is rewritten in the form of RNA is _____.

[Central Dogma]

92. Humans have _____ pairs of chromosomes in total: _____ pairs of autosomes and _____ pair of sex chromosomes.

[Genetics & Heredity]

93. PCR was invented by _____ in _____ and he was awarded a Nobel Prize in _____.

[Recombinant DNA Technology]

94. Caffeine has a half-life of _____ in neonates compared to _____ in adults, illustrating the effect of _____ on drug metabolism.

[Xenobiotic Metabolism]

95. Biliverdin is subsequently converted to _____ by the enzyme _____.

[Heme Degradation]

96. The word 'hormone' is derived from a Greek verb meaning _____.

[Biochemistry of Hormones]

97. In the liver, bilirubin is conjugated with _____ to produce the bile pigment _____.

[Heme Degradation]

98. The fragments of the lagging strand synthesized in short separated segments are called _____.

[DNA Replication]

99. Glands are categorised into _____ (ductless) and _____ glands.

[Biochemistry of Hormones]

100. The Cytochrome P450 isoform responsible for metabolizing over 50% of all clinically prescribed drugs is _____.

[Xenobiotic Metabolism]

101. Steroid hormones are all derivatives of _____ and are _____-soluble.

[Biochemistry of Hormones]

102. The codon for phenylalanine is _____ and the three stop codons are _____.

[Central Dogma]

103. A short fragment of RNA or DNA that must be created before DNA polymerase can begin synthesis is called a _____.

[DNA Replication]

104. DNA polymerase synthesizes new DNA strands by adding nucleotides to the _____ end of the growing strand.

[DNA Replication]

105. Glutathione is a tripeptide composed of _____, _____, and _____.

[Xenobiotic Metabolism]

106. The DNA of chromosomes is tightly coiled around proteins called _____ which support its structure.

[Genetics & Heredity]

107. Cytosine + Ribose = _____

[Nucleic Acid Chemistry]

108. In PCR, the three temperature-dependent steps per cycle are: separation at _____ degrees C, annealing at _____ degrees C, and extension at _____ degrees C.

[Recombinant DNA Technology]

109. Give 3 examples of fat-soluble (steroid) hormones.

[Biochemistry of Hormones]

110. The nitrogenous base found only in DNA (not RNA) is _____, while the base found only in RNA is _____.

[Nucleic Acid Chemistry]

111. A hormone is a _____.

[Biochemistry of Hormones]

112. Alternate forms of a gene located at the same chromosomal locus are called _____.

[Genetics & Heredity]

113. _____ degrades heme into _____, _____, and _____.

[Heme Degradation]

114. Adenine and Guanine are classified as _____ while Cytosine, Thymine and Uracil are classified as _____.

[Nucleic Acid Chemistry]

115. Rh incompatibility during pregnancy can cause a serious condition in the newborn called _____.

[Genetics & Heredity]

116. The melting temperature (T_m) of DNA is the temperature at which _____.

[Nucleic Acid Chemistry]

117. The leading strand undergoes _____ replication while the lagging strand undergoes _____ replication.

[DNA Replication]

118. Substances capable of causing changes in the DNA base sequence are called _____.

[DNA Repair & Mutations]

119. Ataxia Telangiectasia is caused by a defect in the _____ gene, which is involved in _____.

[DNA Repair & Mutations]

120. The isoform of heme oxygenase found mainly in the brain and testes and continuously expressed is _____.

[Heme Degradation]

121. Adenine + Ribose = _____

[Nucleic Acid Chemistry]

122. The N-glycosidic bond in nucleosides joins nitrogen-_____ of purines or nitrogen-_____ of pyrimidines to carbon-_____ of the pentose sugar.

[Nucleic Acid Chemistry]

123. IP3 liberates _____ ions from the mitochondria and endoplasmic reticulum, which then acts as a _____ messenger via calmodulin.

[Biochemistry of Hormones]

ANSWER KEY

Answers are keyed to the shuffled question numbers above.

1. Annealing [Nucleic Acid Chemistry]
2. Porphyrin; iron [Heme Degradation]
3. Vinyl chloride [Xenobiotic Metabolism]
4. Peptide hormones [Biochemistry of Hormones]
5. Messenger RNA (mRNA); ribosomal RNA (rRNA); transfer RNA (tRNA) [Central Dogma]
6. Erwin Chargaff [Nucleic Acid Chemistry]
7. Serum albumin; insoluble [Heme Degradation]
8. Hydroxylation; mono-oxygenases (Cytochrome P450) [Xenobiotic Metabolism]
9. Glucuronidation; acetylation; glutathione conjugation; glycine conjugation (also: sulfation, methylation, water conjugation) [Xenobiotic Metabolism]
10. Receptor molecules embedded in the cell membrane; a second messenger [Biochemistry of Hormones]
11. Cyclic AMP (cAMP) [Biochemistry of Hormones]
12. Helicase [DNA Replication]
13. Helicases [DNA Replication]
14. Hormone response element (HRE) [Biochemistry of Hormones]
15. Carcinogens; carcinogens [DNA Repair & Mutations]
16. Genetic engineering [Recombinant DNA Technology]
17. Carriers; ribosomes [Central Dogma]
18. p arm; q arm [Genetics & Heredity]
19. Lipophilic; non-polar; toxic [Xenobiotic Metabolism]
20. N-acetylcysteine [Xenobiotic Metabolism]
21. Genetic code [Central Dogma]
22. Template strand (antisense strand) [Central Dogma]
23. Degenerate and unambiguous [Central Dogma]
24. Bloodstream [Biochemistry of Hormones]
25. Recessive [DNA Repair & Mutations]
26. One (1); 23 [Genetics & Heredity]
27. Chromosomes [Nucleic Acid Chemistry]
28. Nucleus (nuclei); mitochondria [Nucleic Acid Chemistry]
29. Jaundice [Heme Degradation]
30. mRNA (messenger RNA) [Recombinant DNA Technology]
31. Each daughter DNA molecule retains one original (parental) strand and one newly synthesized strand [DNA Replication]
32. Induction [Xenobiotic Metabolism]
33. 1 million (10^6) [Recombinant DNA Technology]
34. Inositol triphosphate (IP₃); diacylglycerol (DAG) [Biochemistry of Hormones]
35. Urobilin; stercobilin [Heme Degradation]
36. Reverse transcriptase; RNA-dependent DNA polymerase [Recombinant DNA Technology]

37. Cannot perform eukaryotic post-translational modifications; ethical/regulatory concerns [Recombinant DNA Technology]
38. Ferritin [Heme Degradation]
39. Base sequence; replication; DNA repair [DNA Repair & Mutations]
40. Strict specificity of biological action; high biological activity; remote action [Biochemistry of Hormones]
41. Cancer; mutagens [DNA Repair & Mutations]
42. 450 nm; carbon monoxide (CO) [Xenobiotic Metabolism]
43. Bacterium/organism (first letters of genus and species name) [Recombinant DNA Technology]
44. cDNA (complementary DNA) library [Recombinant DNA Technology]
45. Uracil (U) [Central Dogma]
46. Alpha (α)-subunit; cAMP [Biochemistry of Hormones]
47. DNA Polymerase [DNA Replication]
48. Reverse transcriptase [Central Dogma]
49. 1968; genetic code [Central Dogma]
50. Origin of replication [DNA Replication]
51. Thermocycler (or Tempcycler) [Recombinant DNA Technology]
52. Liver; GIT; lungs; skin; kidneys (any 2 others) [Xenobiotic Metabolism]
53. Homozygous; heterozygous [Genetics & Heredity]
54. Foreign [Xenobiotic Metabolism]
55. dCDP (deoxycytidine diphosphate) [Nucleic Acid Chemistry]
56. Replication; transcription; translation [Central Dogma]
57. Heme oxygenase (HO / HO-1) [Heme Degradation]
58. Peptide hormones; steroid hormones; amine (amino acid derivative) hormones [Biochemistry of Hormones]
59. Guanosine; deoxyadenosine [Nucleic Acid Chemistry]
60. Nucleases (endonucleases/restriction endonucleases) [Recombinant DNA Technology]
61. Locus (gene locus) [Genetics & Heredity]
62. Vectors (cloning vectors) [Recombinant DNA Technology]
63. A nitrogenous base; a pentose sugar; a phosphate group [Nucleic Acid Chemistry]
64. Ribozymes [Nucleic Acid Chemistry]
65. DNA Ligase [DNA Replication]
66. Bacteriophages can accommodate larger DNA inserts, making them suitable for cloning large genomic sequences [Recombinant DNA Technology]
67. Nucleotide excision repair [DNA Repair & Mutations]
68. Nucleotides [Nucleic Acid Chemistry]
69. Glutathione [Xenobiotic Metabolism]
70. Two (2); three (3) [Nucleic Acid Chemistry]
71. RNA; reverse transcription [Central Dogma]
72. Ribosomes [Central Dogma]
73. Uridine [Nucleic Acid Chemistry]
74. Xeroderma Pigmentosum (XP) [DNA Repair & Mutations]
75. Cloning vectors; molecular vectors [Recombinant DNA Technology]
76. Electroporation [Recombinant DNA Technology]

77. D-ribose; 2-deoxy D-ribose [Nucleic Acid Chemistry]
78. DNA Ligase [Recombinant DNA Technology]
79. First-pass effect [Xenobiotic Metabolism]
80. Phenotype; genotype [Genetics & Heredity]
81. The heme complex (iron-porphyrin group) [Heme Degradation]
82. -OH (hydroxyl); -NH₂ (amino); -SH (sulfhydryl) [Xenobiotic Metabolism]
83. Adenylate cyclase [Nucleic Acid Chemistry]
84. 32 kDa; 288 amino acid residues [Heme Degradation]
85. Taq Polymerase (derived from *Thermus aquaticus*) [Recombinant DNA Technology]
86. NADPH; NADH [Xenobiotic Metabolism]
87. Nucleoid [Nucleic Acid Chemistry]
88. Nucleotides; phosphodiester bonds; polynucleotides [Nucleic Acid Chemistry]
89. O negative (O⁻); AB positive (AB⁺) [Genetics & Heredity]
90. UDP-glucuronic acid; microsomes [Xenobiotic Metabolism]
91. Transcription [Central Dogma]
92. 23; 22; 1 [Genetics & Heredity]
93. Kary Mullis; 1989; 1993 [Recombinant DNA Technology]
94. 4 days; 4 hours; age [Xenobiotic Metabolism]
95. Bilirubin; biliverdin reductase [Heme Degradation]
96. To excite [Biochemistry of Hormones]
97. Glucuronate (glucuronic acid); bilirubin diglucuronide [Heme Degradation]
98. Okazaki fragments [DNA Replication]
99. Endocrine; exocrine [Biochemistry of Hormones]
100. CYP3A4 [Xenobiotic Metabolism]
101. Cholesterol; fat (lipid) [Biochemistry of Hormones]
102. UUU; UGA, UAA, UAG [Central Dogma]
103. Primer [DNA Replication]
104. 3' (three-prime) end [DNA Replication]
105. Glutamic acid (gamma-glutamyl); cysteine; glycine [Xenobiotic Metabolism]
106. Histones [Genetics & Heredity]
107. Cytidine [Nucleic Acid Chemistry]
108. 95; 50; 72 [Recombinant DNA Technology]
109. Adrenal cortical hormones; androgens; estrogens (any 3) [Biochemistry of Hormones]
110. Thymine; Uracil [Nucleic Acid Chemistry]
111. Chemical messenger secreted in trace amounts by one tissue and carried by the blood to stimulate a specific biochemical or physiological activity in a target tissue [Biochemistry of Hormones]
112. Alleles [Genetics & Heredity]
113. Heme oxygenase (HO-1); biliverdin; carbon monoxide (CO); ferrous iron Fe(II) [Heme Degradation]
114. Purines; pyrimidines [Nucleic Acid Chemistry]
115. Haemolytic disease of the newborn (HDN) [Genetics & Heredity]
116. Half of the helical structure is denatured [Nucleic Acid Chemistry]
117. Continuous; discontinuous [DNA Replication]

118. Mutagens [DNA Repair & Mutations]

119. ATM gene; recognition and repair of damaged DNA [DNA Repair & Mutations]

120. HO-2 [Heme Degradation]

121. Adenosine [Nucleic Acid Chemistry]

122. N-9 of purines; N-1 of pyrimidines; C-1 of the pentose sugar [Nucleic Acid Chemistry]

123. Calcium (Ca²⁺) ions; third messenger [Biochemistry of Hormones]

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